

**fMRI can be highly reliable, but it depends on what you measure**

Philip A. Kragel<sup>1\*</sup>  
Xiaochun Han<sup>2</sup>  
Thomas E. Kraynak<sup>3</sup>  
Peter J. Gianaros<sup>3</sup>  
Tor D. Wager<sup>2\*</sup>

<sup>1</sup> Emory University  
<sup>2</sup> Dartmouth College  
<sup>3</sup> University of Pittsburgh

Running Head: RELIABLE FMRI

\* Please address correspondence to:

Philip A. Kragel  
Department of Psychology  
Emory University  
36 Eagle Row, Atlanta, GA 30322

Tor D. Wager  
Diana L. Taylor Distinguished Professor  
Presidential Cluster in Neuroscience and  
Department of Psychological and Brain Sciences, HB 6207  
Dartmouth College  
Hanover, NH 03755  
603-646-2196

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### **fMRI can be highly reliable, but it depends on what you measure**

In a recent article, Elliot and colleagues (2020) systematically evaluated the reliability of individual differences in task-based fMRI activity, and found reliability to be poor. They conclude: “commonly used task-fMRI measures generally do not have the test-retest reliability necessary for biomarker discovery or brain–behavior mapping.” This is an important and timely effort, and we applaud Elliot et al. for spotlighting the need to evaluate and improve the psychometric properties of fMRI. Larger sample sizes combined with pattern recognition techniques have made translational applications seem finally within reach (Woo et al., 2017). As the field gets serious about using the brain to predict behavior and health outcomes, reliability and other measurement properties will become increasingly important.

However, along with their findings and constructive criticism comes the potential for counterproductive overgeneralization. Though Elliot et al. focus on the oldest and arguably most limited type of fMRI activity measure—average activity levels within isolated brain regions—this paper has already garnered media attention that dramatically mischaracterizes the conclusions drawn from this paper. One headline reads, “...every brain activity study you’ve ever read is wrong”<sup>1</sup>. The sociocultural causes of anti-fMRI (and anti-science) sentiments are not our concern here, but it is necessary to respond to Elliot et al. by pointing out some of the boundary conditions of their critique. fMRI can, in fact, be highly reliable if the right measures are used. In addition, many use cases for biomarkers do not require reliable individual differences.

Reliability estimates in Elliot et al.’s study are handicapped in several ways. Most importantly, they limit their analysis to activity in isolated brain regions. Single-region activity and measures typically have modest effect sizes and predictive validity (Poldrack et al., 2017). However, multivariate measures optimized using machine learning can have dramatically larger effect sizes (Reddan et al., 2017) and reliable individual differences in the “good” to “excellent” range (Gianaros et al., 2020; Woo & Wager, 2016). We discuss several examples below and illustrate them in Figure 1.

In addition to this limitation, the studies in Elliot et al.’s analysis have (a) small sample sizes, (b) little data per participant (often as little as 5 minutes) which substantially reduces reliability; (c) single-task measures rather than the composite-task measures typical in other fields, which can

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<sup>1</sup> Cohen, A. “Duke University researchers say every brain activity study you’ve ever read is wrong”. 6-25-2020 <https://www.fastcompany.com/90520750/duke-university-researchers-say-every-brain-activity-study-youve-ever-read-is-wrong>

be much more reliable (Miyake et al. 2000; Kragel et al. 2018), and (d) long test-retest intervals, up to 140 days in the Human Connectome Project (HCP) data. The latter introduce meaningful changes over time that reduce reliability. Only one study in Elliot et al.'s "unbiased" set had  $N > 50$ , compared with sample sizes  $>1,000$  in a number of modern translational fMRI efforts (e.g., Miller et al. 2016). This introduces noise in the selection of regions that reduces reliability. Elliot et al. acknowledge many of these points, but their cumulative effects paint a far bleaker picture of reliability than the current state-of-the-science affords.

Figure 1 shows four examples of multivariate fMRI measures with high reliability in large samples. Recently, Gianaros et al. (2020) identified two patterns predictive of cardiovascular disease risk using an emotional picture-viewing task and the HCP Emotion task. The test-retest reliability of these multivariate 'signatures' across the first and second half of the session (equivalent to internal consistency) was "good" to "excellent": ( $r_{\text{Spearman-Brown}}$  .82 and .73,  $N = 338$  and 427, respectively; Figure 1a). In contrast, reliability of commonly studied regions (e.g., amygdala) was substantially lower ( $r = .11$  to  $.27$ ). In a second example, we assessed the reliability of Neurologic Pain Signature (NPS) responses—an established neuromarker for pain—in 8 fMRI studies of experimental pain ( $N = 228$ ; data from (Geuter et al., 2020; Jepma et al., 2018)). Reliability was "good" to "excellent" in all studies (Figure 1b). Naturally, reliability may be attenuated over longer time intervals, but brain states may also change meaningfully across time.

Other multivariate fMRI measures also show strong evidence for stable person-level scores across time. For example, Drysdale et al. identified fMRI activity patterns constituting 4 distinct biotypes for depression (total  $N > 1,000$ ). Test-retest analyses across 4 weeks showed excellent (90%) agreement of biotype classifications (Figure 1c). Kamitani and Tong (2005) evaluated the ability to predict the visual angle of a line based on fMRI activity. A multivariate pattern developed during one session predicted visual angle in a second session 31-40 days later with little loss of accuracy (0.7 to 1 degree of error). Together, these examples and many others (e.g., Finn et al., 2015) show that fMRI can yield individual person-level measures that are stable across time.

Finally, biomarkers need not always require high reliability across individuals to be useful. The U.S. FDA (FDA-NIH Biomarker Working Group, 2016) identified 7 biomarker use cases. Some require reliable individual differences (e.g., biomarkers of disease risk and prognostic biomarkers), but others do not (e.g., pharmacodynamic, response, monitoring, and safety

biomarkers). fMRI patterns can robustly detect specific brain states within individuals – e.g., whether a person is viewing a face (Haxby et al., 2001), replaying a memory (Wimmer et al., 2020), paying attention (Rosenberg et al., 2016), engaging in self-regulation (Cosme et al., 2020), or experiencing pain (Duff et al., n.d.). Such measures can be used to monitor the engagement of mind-brain processes, track responses to treatment, and reveal brain mechanisms associated with these. These uses depend on “task reliability,” a measure’s ability to consistently identify a task state in repeated tests (Hedge et al., 2018). We re-analyzed the HCP data presented in Elliot et al., and found that a multivariate model designed to identify engagement of face (vs. shape) processing has “exceptional” task reliability across 4 weeks (Cohen’s  $\kappa = .909$ ; Figure 1d). Thus, even if individual differences are not reliable for a particular task and population, valid and useful biomarkers can be developed.

Ultimately, reliability is not a fixed property of an assay, let alone a whole measurement technology. It depends on details of the tasks, the sample population, and the measures extracted from them (Streiner, 2003). Other recent criticisms of aspects of fMRI (Eklund et al., 2016; Vul et al., 2009) have been used to levy a blanket condemnation of fMRI as a whole. We caution against such overgeneralization, and propose that the findings of Elliot et al. should be considered as a lower bound on the reliability of fMRI. The upper bound of fMRI reliability using state-of-the-art methods is quite high, and remains to be fully explored. We are optimistic that the future is bright for fMRI research focused on biomarker development.

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## Figure Captions

Figure 1. Multivariate brain models can be highly reliable. (a) Reliability of a multivariate signature of cardiovascular disease risk (Gianaros et al., 2020). Inset brain renders depict significant pattern weights fit on brain responses to affective images that positively (warm colors) and negatively (cool colors) contribute to the prediction of a marker of preclinical atherosclerosis. The scatterplot depicts data used to estimate split-half reliability ( $N = 338$ ). (b) Within-session split-half reliability of pain reports, the Neurological Pain Signature (NPS; a multivariate predictive model of physical pain), and average brain activity in pain-related regions of interest for 8 studies that experimentally manipulate pain (Study  $N$ s = 17-50, total  $N = 228$ ; Geuter et al., 2020; Jepma et al. 2018). The intraclass correlation coefficient  $ICC(3, k)$  was computed based on responses to the first and second half of trials. Each circle corresponds to a study. The larger circles and vertical lines indicate the mean and standard error across studies. (c) Agreement of depression biotype classification in two fMRI sessions that were collected 4 weeks apart (Drysdale et al., 2017). Brain renders illustrate nodes whose connectivity was indicative of each biotype relative to healthy controls; red indicates increased connectivity and blue decreased connectivity. (d) Test-retest reliability of classifications in the Emotion task of the Human Connectome Project (Van Essen et al. 2013; Barch et al. 2013). A linear SVM model trained to classify responses to faces and shapes exhibits perfect classification performance ( $AUROC = 1.0$ ) and excellent reliability (Cohen's  $Kappa = .91$ ) in test data from independent subjects.

Figure 1.

